Student Data Smart Contract: Structure, Array and Fallback

Solidity ^0.8.20 | Remix IDE | Deployed on Ethereum Sepolia test network via MetaMask

Step 1: Smart contract code and deployment request

```
pragma solidity ^0.8.20;

contract StudentData {

    struct Student {
        uint256 rollNo;
        string name;
        uint8 age;
        string course;
    }

    Student[] public students;

    event StudentAdded(uint256 indexed rollNo, string name);
    event FallbackCalled(address indexed sender, uint256 value, bytes data);

    function addStudent(
        uint256 _rollNo,
        string memory _name,
        uint8 _age,
        string memory _course
    ) public {
        students.push(Student(_rollNo, _name, _age, _course));
        emit StudentAdded(_rollNo, _name);
    }

    function getStudent(uint256 index)
        public
        view
        returns (uint256 rollNo, string memory name, uint8 age, string memory course)
    {
        require(index < students.length, "Student does not exist");
        Student memory s = students[index];
        return (s.rollNo, s.name, s.age, s.course);
    }

    function getTotalStudents() public view returns (uint256) {
        return students.length;
    }

    function getAllStudents() public view returns (Student[] memory) {
        return students;
    }

    receive() external payable {
        emit FallbackCalled(msg.sender, msg.value, "");
    }

    fallback() external payable {
        emit FallbackCalled(msg.sender, msg.value, msg.data);
    }
}
```

StudentData.sol (comments removed)

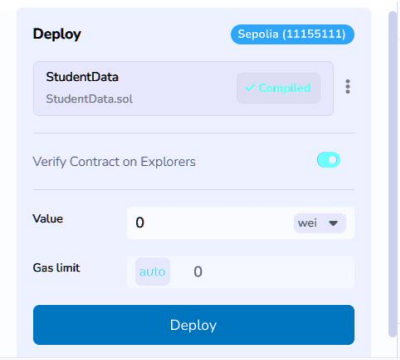


Fig 1: Deploy StudentData on Sepolia (11155111)

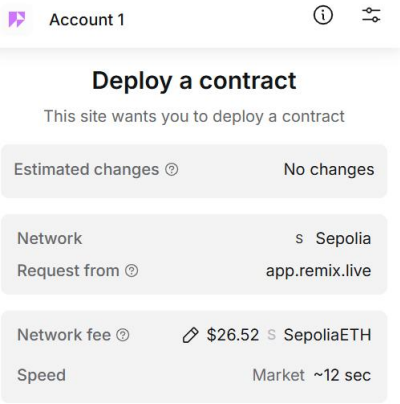


Fig 2: MetaMask deploy request, network Sepolia, fee and speed shown

Step 2: Fallback and receive tested (Remix VM)

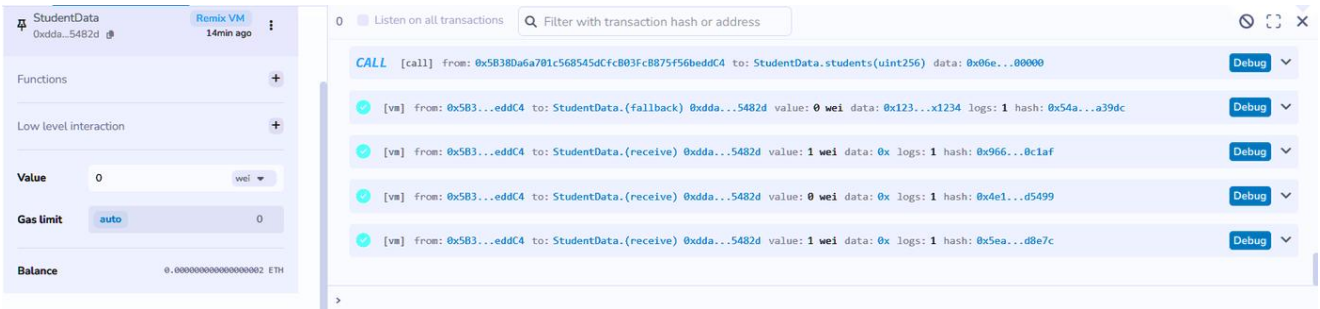


Fig 3: StudentData.(fallback) with calldata 0x1234, and StudentData.(receive) with empty calldata; all succeeded (green) and each emitted 1 log

Step 3: Deployment gas (Remix VM)



Fig 4: Constructor transaction cost 1006120 gas, execution cost 883514 gas

Step 4: Contract deployed on Sepolia

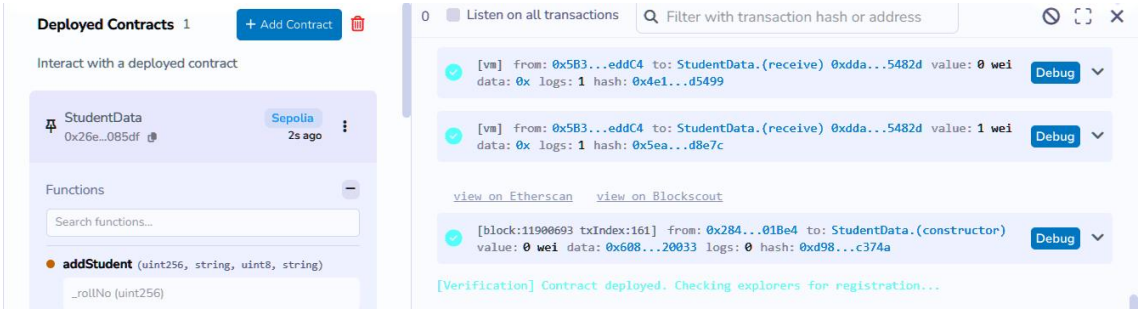


Fig 5: StudentData on Sepolia at 0x26e...085df, constructor in block 11900693

Step 5: addStudent (1, FlankerH, 20, Computer Engg)

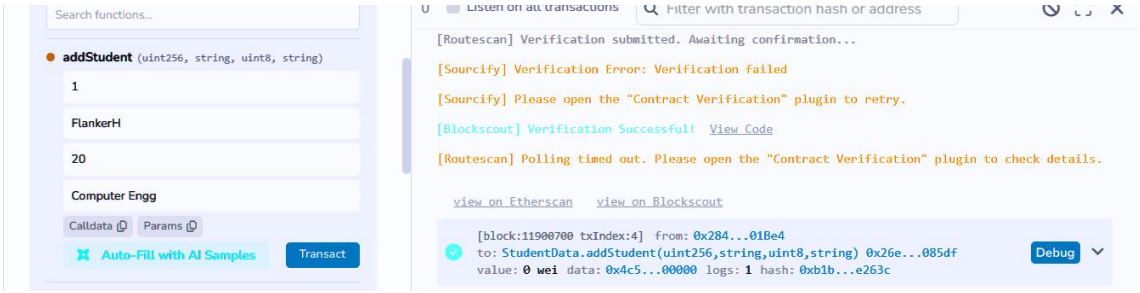


Fig 6: addStudent transaction succeeded on Sepolia, block 11900700, 1 log (StudentAdded event)

Step 6: Reading the stored student data

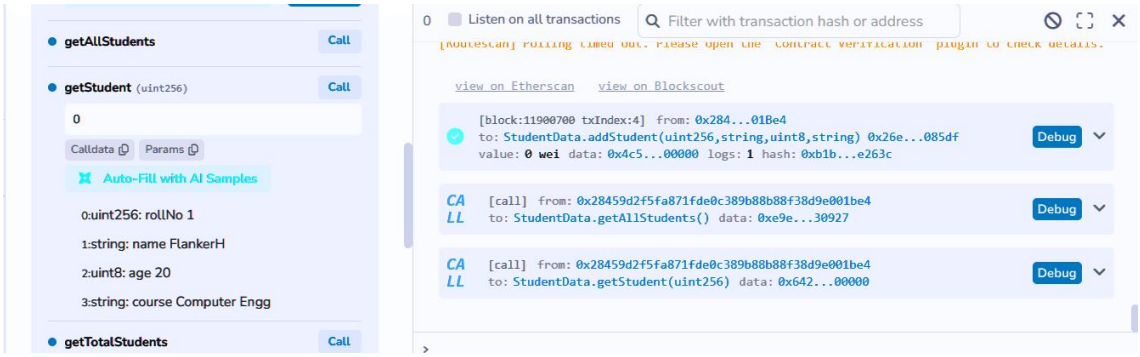


Fig 7: getStudent(0) returns rollNo 1, name FlankerH, age 20, course Computer Engg

Step 7: Transaction fee and gas on Sepolia Etherscan

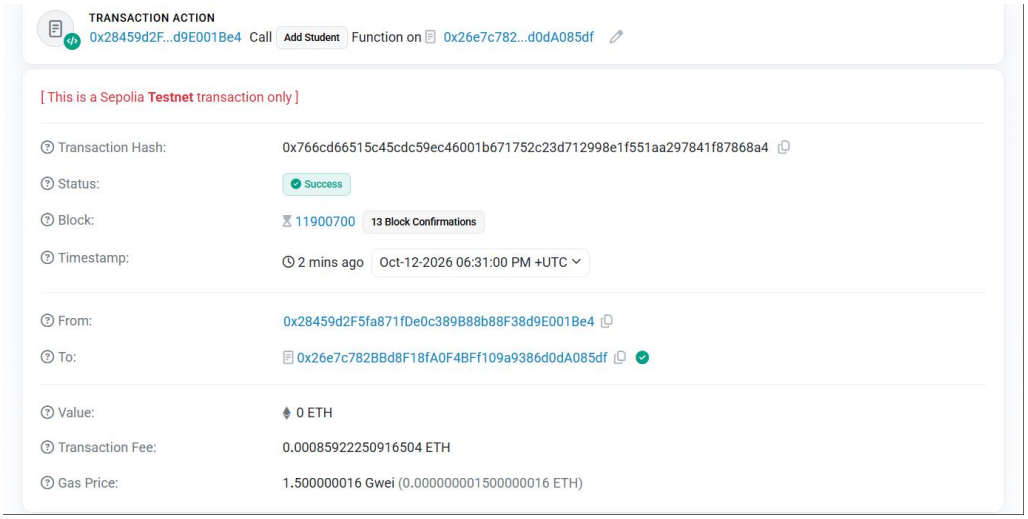


Fig 8: addStudent transaction, Status Success, Transaction Fee 0.00085922250916504 ETH, Gas Price 1.500000016 Gwei

Gas and fee summary

Transaction	Network	Observed
Deploy StudentData	Remix VM	Transaction cost 1,006,120 gas; execution cost 883,514 gas
Deploy StudentData	Sepolia	Success, block 11900693; MetaMask showed a fee estimate (display value only)
addStudent	Sepolia	Fee 0.00085922250916504 ETH at 1.500000016 Gwei, block 11900700
addStudent gas used	Sepolia	572,815 gas (derived: fee / gas price)
fallback (0x1234), receive	Remix VM	Both succeeded; each emitted 1 event log

Conclusion: The contract stores students in a struct inside a dynamic array, and its payable fallback and receive functions accept calls that match no function. Fee = gas used x gas price. Deployment used the most gas (over 1 million on Remix VM), and addStudent used a large amount because it writes to storage.